

MASSFRONT

PLANETS, WORLD SITES, EXPLORATION MODULE, AND COMBINED-ARMS TACTICAL MAP AUDIT

A source-grounded status report with real runtime screenshots, missing-evidence labels, asset inventories, integration risks, and a corrected compact combined-arms level-design contract.

SOURCE	CURRENT EVIDENCE BOUNDARY	DELIVERY STATE
cursor/strip-mass-node-bloom HEAD c4090cc5785e	Shared dirty worktree; reports name exact fingerprints	No publication or activation performed

Prepared 2026-08-25 | Supersedes the incomplete earlier PDF request

1. Executive status

This document distinguishes real runtime content from data-only records, dormant compatibility paths, proposal art, and missing proof. Generated concepts never count as implementation evidence.

Area	Current result	Meaning
War Table planets	PASS	Four distinct authored worlds; zero page/GL/context errors; 10.4 ms catalog render p95; no extra WebGL context.
Base site templates	FAIL - 6/8	Six real captures. Brutalist Prefecture and Pressure Dome Court fail deterministic placement; no screenshots fabricated.
Exploration module	NOT READY	Standalone and Settings-only by design. Host authority, exactly-once results, menu parity, social routing, optional pack, and phone-first evidence remain incomplete.
Combined-arms interiors	DESIGN ONLY	No executable XS/Small tactical map system yet. Corrected spec supports personnel, light mechs, small vehicles, and selected medium mechs.
Release	NOT ACTIVATED	npm run gate passes, but this report does not authorize or perform HF/Cloudflare publication.

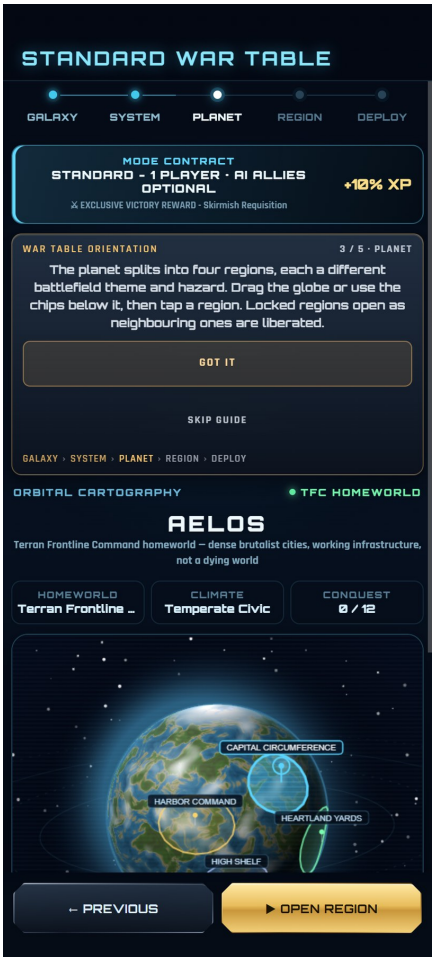
Verified content counts

Base homeworlds	Regions	Advertised maps	Active site templates	Exploration planets	Orbital contacts	Ground-site seeds	NEXUS-VII districts
4	16	48	8	6	9	9	11

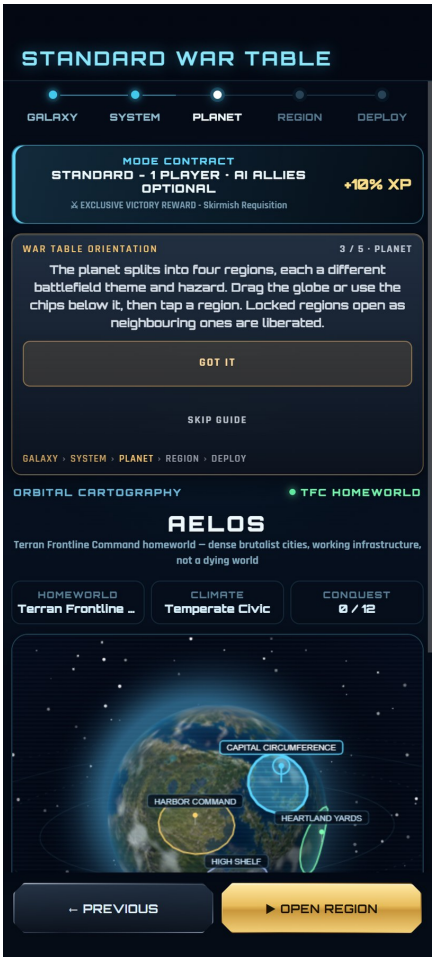
- ☐ No dedicated neutral military_base template exists; military locations are currently composed from outposts, world-kit structures, and map semantics.
- ☐ The nine Galactic ground sites are mission data envelopes, not playable geometry, cover, navmesh, or interior levels.
- ☐ The 16 existing _small maps are compact outdoor RTS theatres on the normal strategic map system, not XCOM-scale missions.
- ☐ The full authoring module remains preserved; optional runtime-pack optimization must be derived, reversible, and visually approved.

2. War Table planet conversion - matched evidence

The base renderer now borrows its existing WebGL2 context, renders a bounded authored PBR globe into an FBO, restores GL state, and composites into the existing Canvas2D War Table. Region caps, labels, drag rotation, and hit testing remain in the base path.



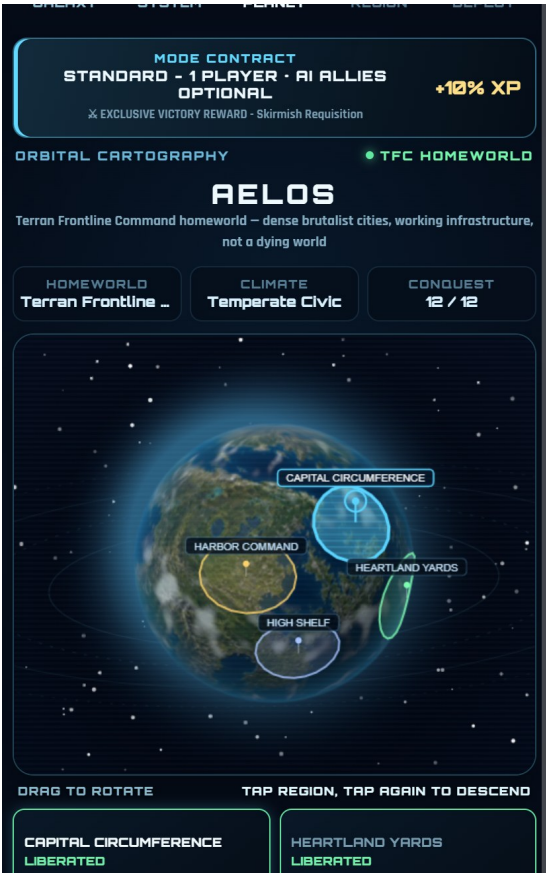
BEFORE - procedural Aelos
dacdbef8...e678d



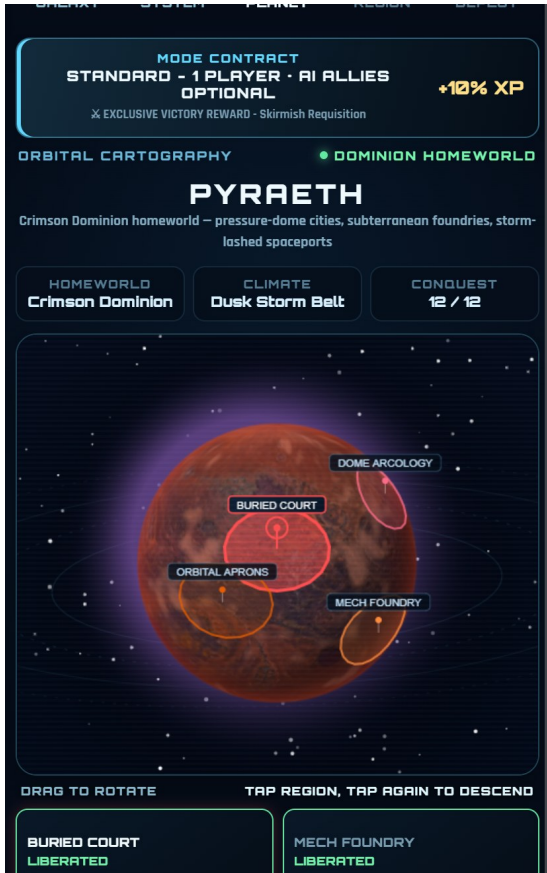
AFTER - authored Aelos
db93d0b...d8bd1c

Hardware: ANGLE/D3D11 on AMD Radeon 610M. Medium tier loaded basecolor, normal, and clouds for every world. Decoded bytes 6,291,456; estimated resident GPU bytes 8,388,608; added contexts 0; GL error 0; context losses 0.

3.1 Authored homeworld identity



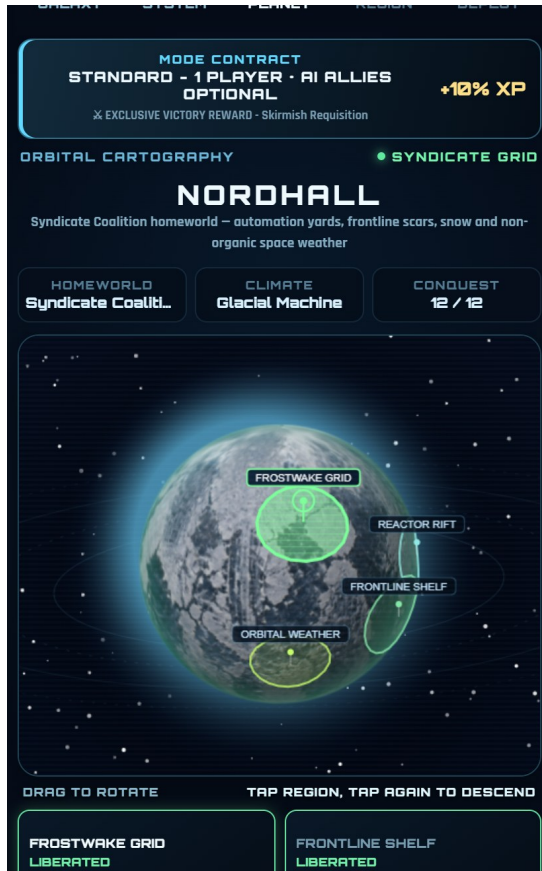
AELOS - temperate civic capital



PYRAETH - dusk storm foundry

These are real packaged-runtime captures. They are not recolors: each planet uses a distinct source family and authored surface structures. Region overlays remain the existing interactive War Table layer.

3.2 Authored homeworld identity



NORDHALL - glacial machine vault



VESPERA - Brood volcanic hive

These are real packaged-runtime captures. They are not recolors: each planet uses a distinct source family and authored surface structures. Region overlays remain the existing interactive War Table layer.

4. AELOS - regions and 12 advertised maps

Region / POI	Theme	Compact	Medium	Large
CAPITAL CIRCUMFERENCE aelos_north POI: Command Circumference	verdant	Parade Circumference aelos_north_small	Civic Grid aelos_north_medium	Frontline Capital aelos_north_large
HEARTLAND YARDS aelos_basin POI: Heartland Foundry	verdant	Greenbelt Works aelos_basin_small	Canal Prefecture aelos_basin_medium	Heartland Yards aelos_basin_large
HARBOR COMMAND aelos_coast POI: Port Admiralty	verdant	Harbor Battery aelos_coast_small	Admiralty Causeway aelos_coast_medium	Pelagic Ring aelos_coast_large
HIGH SHELF aelos_ridge POI: Great Divide Gate	arctic	High Shelf Fort aelos_ridge_small	Stone Rampart aelos_ridge_medium	Divide Gate aelos_ridge_large

World identity: Civic capital, harbor, foundry heartland, and high-shelf fortification. These are runtime campaign records, not a claim that every named POI has a unique model.

4. PYRAETH - regions and 12 advertised maps

Region / POI	Theme	Compact	Medium	Large
BURIED COURT pyraeth_crater POI: Court of Iron	vespera	Crown Pit pyraeth_crater_small	Underforge pyraeth_crater_medium	Buried Court pyraeth_crater_large
MECH FOUNDRY pyraeth_belt POI: Promethean Mega-Grid	vespera	Mech Gate pyraeth_belt_small	Iron Pyre Yards pyraeth_belt_medium	Dominion Works pyraeth_belt_large
DOME ARCOLOGY pyraeth_caldera POI: Ignis Dome Court	vespera	Dome Crown pyraeth_caldera_small	Arcology Crucible pyraeth_caldera_medium	Red Expanse pyraeth_caldera_large
ORBITAL APRONS pyraeth_flats POI: Hub Delta Pads	vespera	Apron Mile pyraeth_flats_small	Blackwind Ports pyraeth_flats_medium	Cinder March pyraeth_flats_large

World identity: Pressure domes, subterranean foundries, crater infrastructure, and orbital aprons. These are runtime campaign records, not a claim that every named POI has a unique model.

4. NORDHALL - regions and 12 advertised maps

Region / POI	Theme	Compact	Medium	Large
FROSTWAKE GRID nordhall_isles POI: Archipelago Core Vault	arctic	Frostwake Docks nordhall_isles_small	Rimewater Chain nordhall_isles_medium	Boreal Grid nordhall_isles_large
FRONTLINE SHELF nordhall_cliff POI: Citadel Command Pinnacle	arctic	Scar Approach nordhall_cliff_small	Arcology Steps nordhall_cliff_medium	Citadel Shelf nordhall_cliff_large
REACTOR RIFT nordhall_frost POI: Pale Trench Reactor	arctic	Cryo Rift nordhall_frost_small	Glacier Fault nordhall_frost_medium	Pale Chasm nordhall_frost_large
ORBITAL WEATHER nordhall_peaks POI: Skyshield Array	arctic	Valkyrie Spur nordhall_peaks_small	Skyshield Range nordhall_peaks_medium	Crown of Winter nordhall_peaks_large

World identity: Glacial automation, reactor rifts, fortified shelves, and weather arrays. These are runtime campaign records, not a claim that every named POI has a unique model.

4. VESPERA - regions and 12 advertised maps

Region / POI	Theme	Compact	Medium	Large
CALDERA NESTS vespera_spire POI: Great Hive Spire	ashland	Hatch Pit vespera_spire_small	Spire Brood vespera_spire_medium	Sol Nest Crown vespera_spire_large
INFESTATION FIELDS vespera_dunes POI: Tide Relay Net	ashland	Ichor Narrows vespera_dunes_small	Sable Bloom vespera_dunes_medium	Eventide Tide vespera_dunes_large
MAGMA HATCHERIES vespera_refinery POI: Megaforge Spine	ashland	Magma Crypt vespera_refinery_small	Derelict Brood vespera_refinery_medium	Silent Megaforge vespera_refinery_large
OVERGROWN FRONT vespera_plateau POI: Terminator Hive Spire	verdant	Twilight Mesa vespera_plateau_small	Gloam Ramparts vespera_plateau_medium	Terminator Front vespera_plateau_large

World identity: Hive spires, infestation fields, dead refineries, magma hatcheries, and overgrown fronts. These are runtime campaign records, not a claim that every named POI has a unique model.

Brood remains a non-playable, non-humanoid Galactic enemy in the exploration design. Base horde identifiers require an explicit adapter rather than silent equivalence.

5. Terrain, structures, and material library

Theme	Identity	Current authored PBR families	Status
Verdant	grass/highland/civic temperate	planet-map-v2 verdant-highland; grass/soil/paving sheets	RUNTIME
Ashland	basalt, soot, burned rock	ashland-basalt location and planet-map-v2 channels	RUNTIME
Arctic	wind-packed snow, ice, rock	arctic-windpack location channels	RUNTIME
Vespera	alien volcanic crust	vespera-crust location channels	RUNTIME

World-kit models

ID	Triangles	Role	Shared runtime material
barracks	1,227	military housing	mf-worldkit-v4
tower	1,262	civic/defensive tower	mf-worldkit-v4
block	645	modular city block	mf-worldkit-v4
depot	1,013	logistics/refinery	mf-worldkit-v4
watchtower	1,184	defensive observation	mf-worldkit-v4
gauss	1,277	relic/weapon emplacement	mf-worldkit-v4
gatehouse	1,185	outpost/city entrance	mf-worldkit-v4

The shared generic city kit also renders tower, dome, hall, tank farm, civic block, and skyline-anchor kinds. Those pieces are compositional assets, not one bespoke model per settlement. Material-file counts must not be reported as unique buildings.

6. Active base site templates - real phone catalog



Current result: FAIL - 6/8 accepted

Template	Class	Capture
Ridge Gate Outpost	outpost	PASS
Supply Yard	outpost	PASS
Gauss Shrine	relic	PASS
Broken Span	relic/ruin	PASS
Walled Town	city/colony	PASS
Brutalist Prefecture	city	BLOCKED
Orbital Apron	spaceport/industrial	PASS
Pressure Dome Court	dome colony	BLOCKED

The two blank panels are intentional. The real planner rejected those templates after eight deterministic seeds; the catalog does not fabricate screenshots.

7. Location-class gaps and planner blockers

Requested content	What exists now	Honest status
Outposts	Two explicit layouts	RUNTIME
Cities/colonies	Two explicit city layouts plus generic city kit	RUNTIME, limited variety; one active city template planner-blocked
Relics/ruins	Two explicit relic layouts plus semantic POIs	RUNTIME templates + semantic names
Spaceports	One Orbital Apron layout	RUNTIME
Pressure domes	One dome-cluster layout	ACTIVE DATA, planner-blocked
Military bases	Outposts, forts, player structures, and mission semantics	Dedicated template/model MISSING
Refineries	Depot/tank-farm kit and Vespera/Pyraeth map semantics	Distinct refinery template MISSING
Derelict megastructures	Exploration contact GLB and ground-site seeds	Space model RUNTIME; tactical map DATA-ONLY

Planner root cause

Both blocked templates place their first required plot on a cross-street intersection. The frontage pass moves the plot beside one street, but the perpendicular street still bisects its 1.18x apron. Because the first plot is required, the full site rolls back. This is deterministic geometry, not bad random seeds.

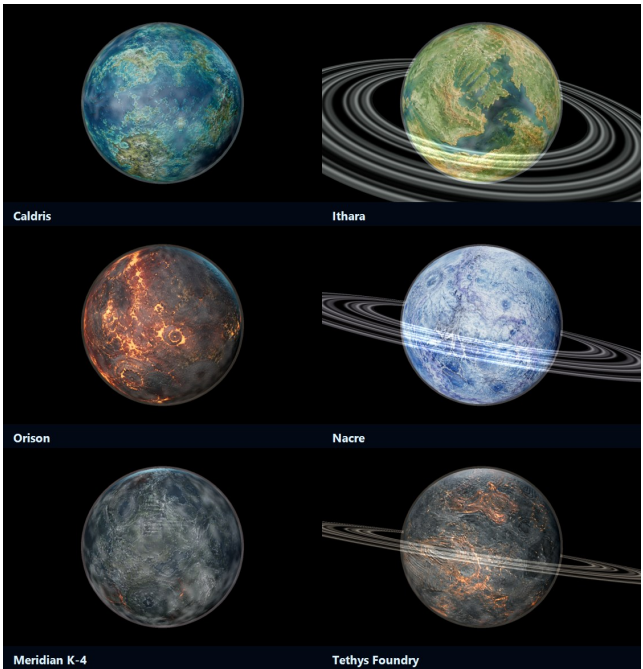
- Gameplay impact: the current caller ignores the failed stamp result, silently dropping 11 Aelos prefecture requests across 10 maps and 11 Pyraeth pressure-dome requests across 7 maps. The smallest safe repair is a template-authored plaza gap; weakening the shared road-clearance rule would hide the symptom and risk other sites.
- Art-quality caution: PASS in the catalog means a valid source-matched capture, not visual approval. The six captured sites reuse the same seven-mesh kit and palette, several roads measure only 9-17 world units (narrower than a TITAN diameter), and the shrine, broken span, town wall, and orbital apron need stronger bespoke silhouettes.
- Recommended repair: author a plaza/intersection gap or move the required anchor off the crossing, then add an exact-template placement probe across the natural applicable maps. Re-run the 8-template phone catalog after the source fingerprint changes.

8. Exploration orbital contacts - nine authored GLB roots



Authored preview contact sheet: embassy spindle, logistics/refinery array, relay gates, archive hulk, quarantined colony spine, and lifeboat debris field.

9. Exploration planets and surface landmarks



Six exploration planets. Each has basecolor, normal, ORM, height, emissive, and cloud channels. Surface discoveries currently use generic survey beacons rather than close-up colony/refinery models.

Planet	Known surface landmarks / discoveries
Caldris	See catalog data
Ithara	See catalog data
Orison	See catalog data
Nacre	See catalog data
Meridian K-4	See catalog data
Tethys Foundry	See catalog data

10. Galactic ground-site seeds - data, not levels

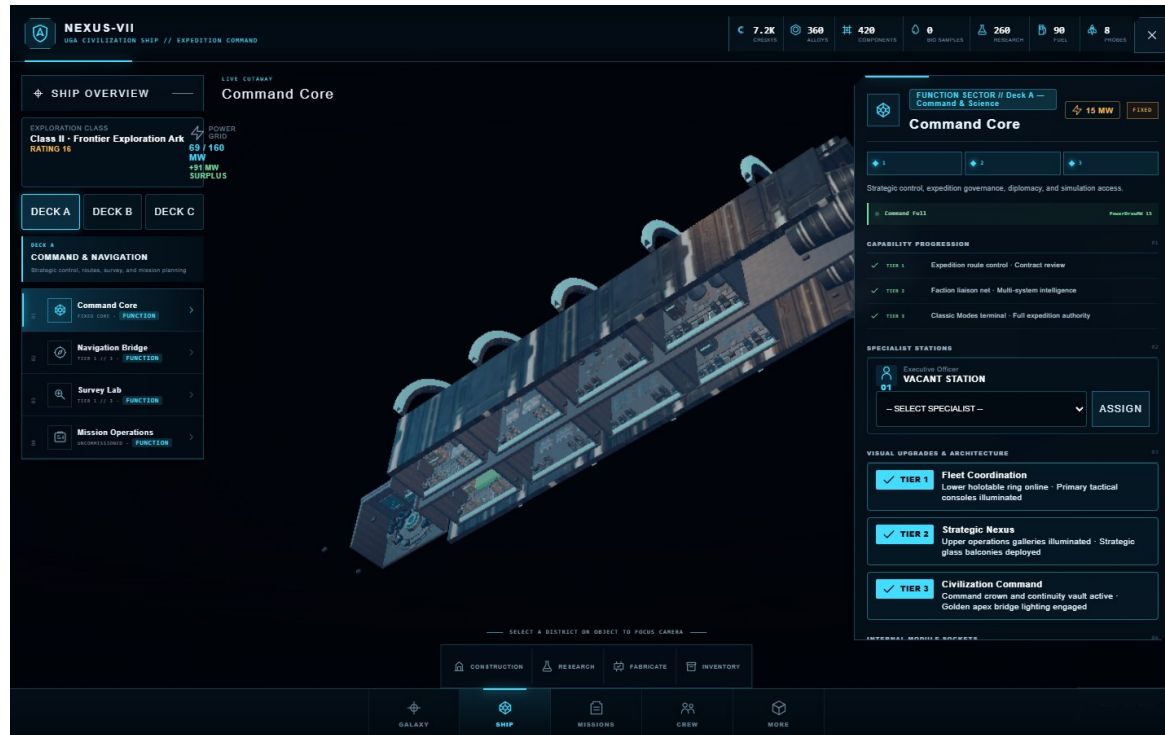
Site ID	Site / mission	Landing zones	Implementation
aelos_caldris	Caldris Orbital Ring		DATA-ONLY - no tactical geometry/navmesh
aelos_heliograph	Heliograph Relay		DATA-ONLY - no tactical geometry/navmesh
aelos_freeport	Morrow Freeport		DATA-ONLY - no tactical geometry/navmesh
veyra_orison	Orison Derelict		DATA-ONLY - no tactical geometry/navmesh
veyra_lens	Lensing Observatory		DATA-ONLY - no tactical geometry/navmesh
veyra_ossuary	Ossuary Vault		DATA-ONLY - no tactical geometry/navmesh
karak_meridian	Meridian Colony		DATA-ONLY - no tactical geometry/navmesh
karak_spine	Colony Transit Spine		DATA-ONLY - no tactical geometry/navmesh
karak_hive	Karak Primary Hive		DATA-ONLY - no tactical geometry/navmesh

Deployment structures

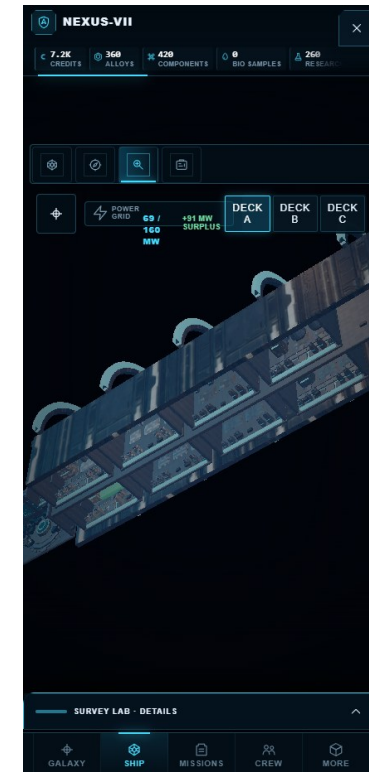
Template	Role	Combined-arms rule
field_relay	communications	portable, anchor-only
resource_processor	economy	exterior/refinery pad only
defensive_emplacement	defense	portable/prebuilt hardpoint with arc and egress checks
forward_command	production	exterior colony/base mission on an authored pad

The current LocalSandboxHost simulates operation results. It does not launch a real tactical map. The integrated host must resolve abstract deployment groups into faction-specific runtime unit IDs, footprints, nav classes, and site compatibility before launch.

11. NEXUS-VII current ship and district evidence



Landscape diagnostic overview



Phone diagnostic overview

These captures are useful design evidence, but the current all-room set is not accepted as a source-matched phone-first matrix. The observed rooms remain box-like, detached, materially repetitive, and too often hidden by mobile inspector chrome.

12. NEXUS-VII - all 11 district captures



Diagnostic contact sheet only. Command, Navigation, Survey, Mission Operations, Research, Fabrication, Engineering, Habitat, Embassy, Strike/Expedition Bay, and Logistics all exist as focusable districts.

13. Exploration integration readiness

The module correctly remains isolated, disabled by default, and reachable only through the Experimental setting when eventually approved. It must not appear as a Galactic Campaign button on the production main menu.

Priority	Gap	Required correction
P0	No production MassfrontIntegratedHost	Async host authority for profile, campaign, transactions, social, audio, pack, Classic, ground launch/result, recovery, and disposal.
P0	Exactly-once result order is unsafe	Apply state and receipt atomically; a crash cannot record receipt before rewards.
P0	Ground sites do not map to playable descriptors	GroundOperationMapDescriptorV1 with dimensions, nav classes, portals, LOS, destruction, and asset hashes.
P0	Duplicated account/progression/faction IDs	Single owner plus explicit ID adapters; Classic cannot mutate Galactic campaign state.
P1	Main-menu/War Table/social parity incomplete	Route base-owned functions through host surfaces; do not duplicate social or import module globals.
P1	Optional pack is oversized and unmanaged	Curated allowlist, compression with matched captures, install/resume/hash/rollback, authoring sources preserved.
P1	Phone evidence stale/incomplete	At least 45% ship visibility, collapsed inspector, 48 px controls, Back/focus/tap safety, current fingerprinted matrix.

Stored readiness remains NOT READY (48 pass, 1 fail, 7 unknown). The latest measured optional runtime subset is 166.06 MiB and lacks an approved pack budget plus source-matched compression evidence.

14. Corrected compact combined-arms design contract

XCOM is the inspiration for roster preparation, mission consequences, objective density, cover, compact pacing, and extraction. MASSFRONT combat remains real-time combined arms. Personnel-only rooms are not the target.

Format	Playable dimensions	Duration	Normal force envelope
XS	192 x 160 m	8-12 min	4-8 personnel, 2-6 light mechs, up to 2 small vehicles; medium mech only in a high-bay variant
Small	320 x 256 m	12-20 min	8-16 personnel, 6-12 light mechs, 2-4 small vehicles, optional 1-2 medium mechs
Medium	480 x 384 m	18-30 min	16-32 personnel, 12-24 light mechs, 4-8 small vehicles, 2-4 medium mechs

Environment element	Recommended minimum
Small-vehicle gate	18 m wide x 8 m high
Medium-mech gate	24 m wide x 12 m high
One-way light lane	16-18 m
Two-way light lane	28-30 m
Medium-mech turning court	42-48 m diameter
Light combined-arms ceiling	8 m
Medium-mech/high-bay ceiling	12-14 m
Light vehicle ramp	6-8%; hard maximum 8.3%

Gameplay blockers are P0: closed doors and walls must block acquisition and shots; breaches must update deterministic nav and LOS; destruction cannot remove the final mandatory route; admitted vehicles must have a complete route, turning court, and extraction path.

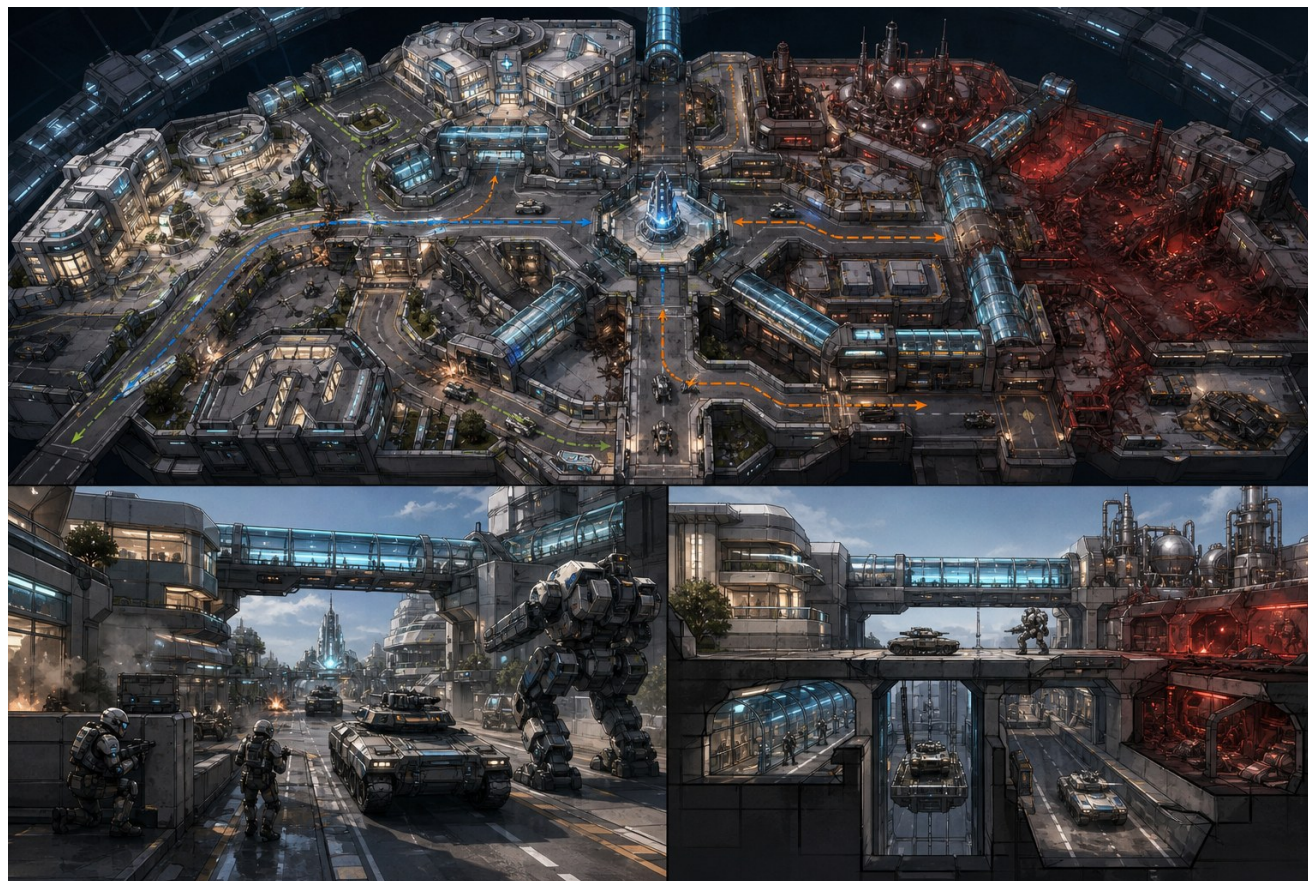
15. XS transit deck concept - mood accepted, layout unapproved



CONCEPT - NOT RUNTIME. Directionally correct combined-arms scale: light vehicle loop, compact mech bay, personnel flank, breach damage, high-bay section, and shared location across overview/keyframe/section.

Approval still requires: scale bar, route/nav-class overlay, exact gate/turn/ramp dimensions, deployment volumes, cover/LOS graph, destructible-state diagram, and runtime silhouettes.

16. Small colony spine concept - mood accepted, layout unapproved



CONCEPT - NOT RUNTIME. Directionally correct hierarchy: colony roads, glass tunnels, light armor, medium mech, vertical freight movement, inhabited and quarantined sectors, and a central operation objective.

The earlier two infantry-room boards are rejected and intentionally excluded from this report's approved direction.

17. Tactical mission families and implementation order

Mission family	Size	Combined-arms level identity
Derelict archive boarding	XS/Small	Docking high bay, broken transit spine, vehicle loop, personnel maintenance shortcuts, breachable archive wall
Relay seizure/defense	XS/Small	Exterior apron plus interior service lanes; armor holds apron while personnel reach controls
Quarantine colony	Small/Medium	Road loop, clinic, service tunnels, landing court, Brood nests, civilian evacuation
Refinery restart/sabotage	Small/Medium	Vehicle circulation around process vessels; overhead personnel routes and pressure hazards
Military-base breach	Small/Medium	Outer gate, motor pool, two breach routes, command core, anchored support pads
Ship-to-ship docking	XS	Cargo lock, vehicle elevator, hangar spine, personnel service branches; no space combat
Brood hive depth	Small/Medium	Mech-sized main artery, personnel crawl routes, membrane gates, acid hazards

Implementation sequence

- ☐ Freeze GroundOperationMapDescriptorV1 and measured clearance classes.
- ☐ Boot deterministic descriptor-driven maps without changing Classic map allocation.
- ☐ Add class-aware navigation, portal graph, gameplay LOS/shot blockers, and destruction states.
- ☐ Resolve abstract Strike Team choices to faction-specific runtime IDs and anchored deployables.
- ☐ Build one vertical slice first: Orison Derelict or Colony Transit Spine, phone-first.
- ☐ Expand only after runtime movement, formation, LOS, destruction, camera, and S25 evidence pass.
- ☐ Enable MassfrontIntegratedHost in developer builds; keep the public experiment Settings-only and off by default.

18. Whole-plan priority order and evidence ledger

P0 - architectural blockers

- ☐ S25 Ultra accepted evidence lane before 500-1000-unit performance claims.
- ☐ Combined-arms descriptor, navigation classes, LOS/shot blockers, destruction/portal states, and deployment authority.
- ☐ Exploration host authority, atomic exactly-once operation return, ID adapters, and Settings-only experimental routing.
- ☐ Fix the two active site-template placement blockers and repeat the source-matched catalog.

P1 - quality and coherence

- ☐ Connected NEXUS-VII multi-deck architecture, differentiated interior materials, and phone room visibility.
- ☐ Menu, War Table, progression, research/crafting, social, audio, update, and diagnostics parity through host adapters.
- ☐ Authored military-base, refinery, colony, derelict, and extraction-complex templates with runtime captures.
- ☐ Core mobile selection, production/research clarity, movement, tap safety, and full interface audit.

Evidence files

Evidence	Result / SHA-256 prefix
tmp/wartable-planet-conversion/after/evidence.json	PASS / 9d962f39870c
tmp/wartable-planet-conversion/after/catalog-phone-412x915-all-worlds-contact-sheet.png	PASS / 4f6193fd25a2
tmp/site-template-catalog/run-2026-08-26T01-16-54-914Z/report.json	FAIL 6/8 / d4f0db3082fd
tmp/site-template-catalog/run-2026-08-26T01-16-54-914Z/active-site-templates-phone-contact-sheet.png	6 captures + 2 honest blockers / ea8619057e8a
tmp/combined-arms-map-audit/COMBINED_ARMS_TACTICAL_MAP_SPEC_2026-08-25.md	design audit / c537c27c0bdc
npm run gate	PASS: global scope, 91-source bundle, 136.7 MiB staged www

Bottom line

The four base planets are now a real, verified improvement. The content library is meaningful but uneven: six base templates have current captures, two active templates are placement-broken, dedicated military-base/refinery classes are missing, and Galactic ground sites remain data-only. The next high-value milestone is one measured, playable combined-arms operation linked through a safe experimental host - not more unverified room concepts.